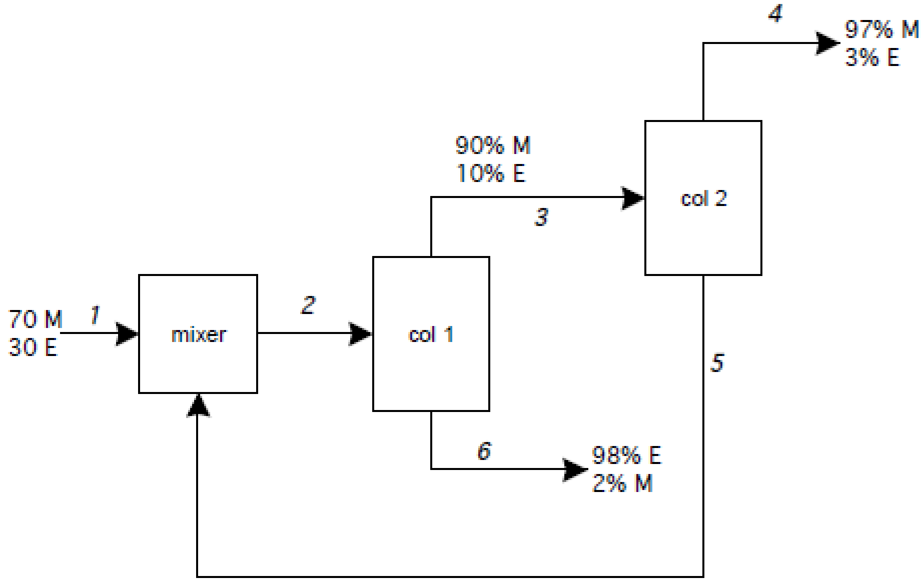
P5.32

The flow diagram is shown, with E for ethane and M for methane.



Flows are given in gmol/min, compositions in mol%, and we assume steady-state operation. We write material balances on all 3 process units. We will use mole fractions and total molar flow rates as variables, because of the nature of the information provided.

Mixer:

$$70 + x_{M5}(100) = x_{M2}\dot{n}_2$$
$$30 + x_{E5}(100) = x_{E2}\dot{n}_2$$

Column 1:

$$x_{M2}\dot{n}_2 = 0.9\dot{n}_3 + 0.02\dot{n}_6$$
$$x_{E2}\dot{n}_2 = 0.1\dot{n}_3 + 0.98\dot{n}_6$$

Column 2:

$$0.90\dot{n}_3 = 0.97\dot{n}_4 + x_{M5}(100)$$
$$0.10\dot{n}_3 = 0.03\dot{n}_4 + x_{E5}(100)$$

We also have

$$x_{M5} + x_{E5} = 1$$
$$x_{M2} + x_{E2} = 1$$

We have 8 equations in 8 unknowns. We can proceed by simultaneously solving all 8 equations (using an equation-solving package such as EES), or by setting up the matrices as we learned in Chapter 3 and solving using a matrix function on a calculator.

Alternatively, we can first group all 3 units into a single system and derive 2 equations in 2 unknowns:

$$70 = 0.97\dot{n}_4 + 0.02\dot{n}_6$$
$$30 = 0.03\dot{n}_4 + 0.98\dot{n}_6$$

(or, we use one of these balances plus a total mole balance equation). We solve by substitution and elimination to find

$$\dot{n}_4 = 71.6 \text{ gmol/min}$$
$$\dot{n}_6 = 28.4 \text{ gmol/min}$$

From a total mole balance around column 2, we find

$$\dot{n}_3 = \dot{n}_4 + \dot{n}_5 = 71.6 + 100 = 171.6 \text{ gmol/min}$$

This result inserted into the component balance equations on column 2 tells us

$$x_{M5} = 0.85$$
$$x_{E5} = 0.15$$

Now we move backwards to column 1 balances:

$$x_{M2}\dot{n}_2 = 0.9(171.6) + 0.02(28.4) = 155 \text{ gmol/min}$$
$$x_{E2}\dot{n}_2 = 0.1(171.6) + 0.98(28.4) = 45 \text{ gmol/min}$$

or

$$\dot{n}_2 = 200 \text{ gmol/min}$$
$$x_{M2} = 0.775$$
$$x_{E2} = 0.225$$

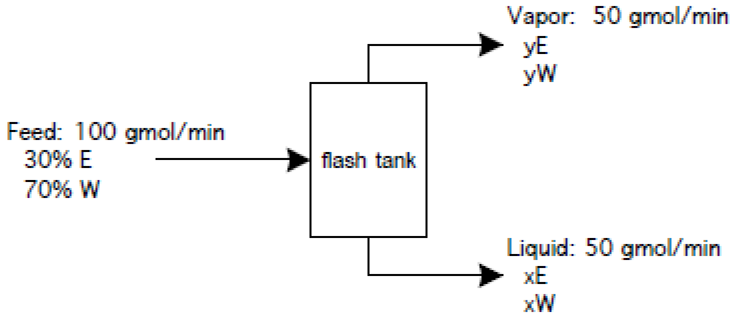
The flowrate of the desired product, stream 4, is 71.6 gmol/min. The recycled stream, stream 5, is 85% methane and 15% ethane. The fractional recoveries of methane and ethane are:

$$f_{RM4} = \frac{x_{M4}\dot{n}_4}{x_{M1}\dot{n}_1} = \frac{0.97(71.6)}{0.70(100)} = 0.992$$

$$f_{RE6} = \frac{x_{E6}\dot{n}_6}{x_{E1}\dot{n}_1} = \frac{0.98(28.4)}{0.30(100)} = 0.928$$

P5.34

Assuming a basis of 100 gmol/min, the flow diagram is



where we've used E for ethanol and W for water; compositions are given in mole fractions.

The material balance equations are

$$0.3(100) = 30 = y_E(50) + x_E(50)$$

$$0.7(100) = 70 = y_W(50) + x_W(50)$$

We also know of course that

$$y_E + y_W = 1.0$$

$$x_E + x_W = 1.0$$

Figure 5.13 (and Table B.11) provides the phase equilibrium relationships (relating y_E to x_E, and y_W to x_W). Since the information is in graphical rather than equation form, we are stuck with a trial-and-error approach: guess a temperature, find y_E and x_E from the graph, insert values into the material balance equation and see if the equation is satisfied.

Try $T = 84.1$°C: $y_E = 0.5089$ and $x_E = 0.1661$ (Table B.11). In material balance;

$$y_E(50) + x_E(50) = 0.5089(50) + 0.1661(50) = 33.75 \neq 30$$

Try $T = 85.3$°C: $y_E = 0.4704$ and $x_E = 0.1238$ (Table B.11). In material balance;

$$y_E(50) + x_E(50) = 0.4704(50) + 0.1238(50) = 29.7 \sim 30$$

This is close enough – a temperature of about 85°C will vaporize 50% of the feed stream and produce a vapor of 47 mol% ethanol and a liquid of 12.4 mol% ethanol.

The fractional recoveries and separation factors are:

$$f_{REv} = \frac{y_E \dot{n}_v}{z_{Ef} \dot{n}_f} = \frac{0.47(50)}{0.30(100)} = 0.78$$

$$f_{RWl} = \frac{x_W \dot{n}_l}{z_{Wf} \dot{n}_f} = \frac{0.876(50)}{0.70(100)} = 0.626$$

$$\alpha_{E-W} = \frac{y_E}{x_E}\frac{x_W}{y_W} = \left(\frac{0.47}{0.124}\right)\left(\frac{0.876}{0.53}\right) = 6.26$$